Run report: plain XL at degree 5 on the m·ℓ = 21 cells

Study `research/dreg_xl_degree5_20261018/`; preregistered in commit 88451b3e2. **Run** run-20261018T2104Z (queue log 2026-10-18T21:04:12Z to 22:10:30Z). **Class:** stage diagnostic; no S, no rho ratio, no speed claim.

What ran

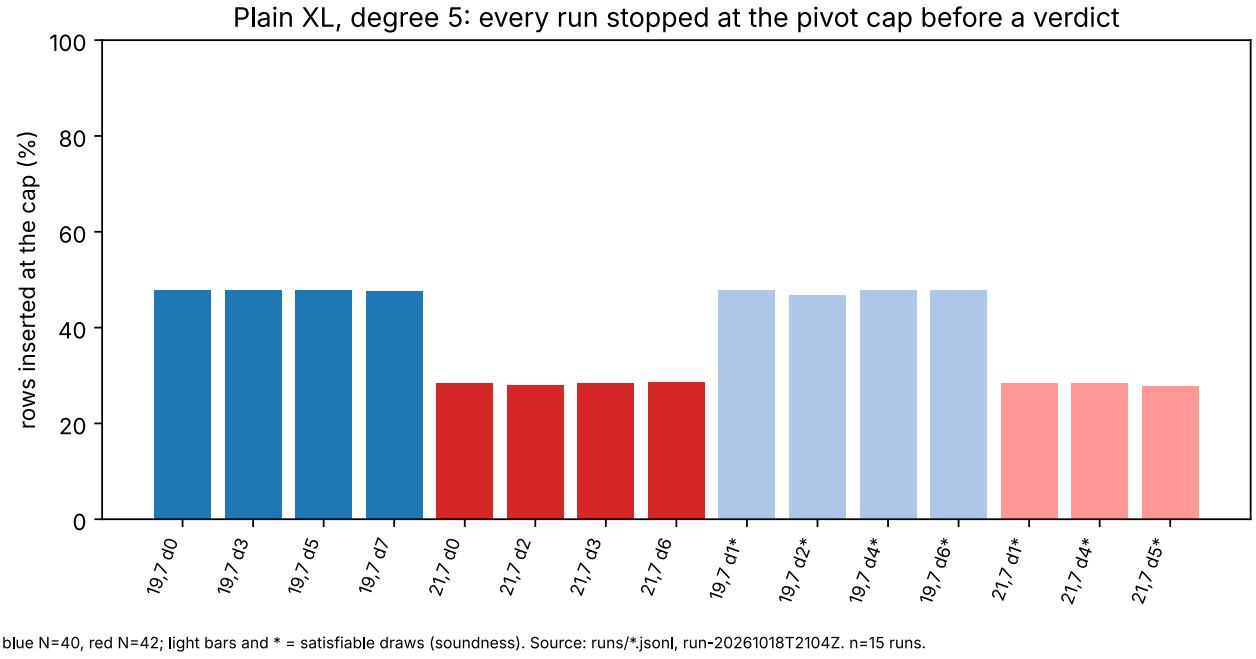
- Command: `run.sh registered`, one process per cell, `prlimit --as=8000000000 timeout 10800`, `RAYON_NUM_THREADS=2`, engine `examples/xl_dense_degree run --degree 5 --threads 2 --max-bytes 6300000000`.
- Draws: unsatisfiable draws 0, 3, 5, 7 of m3 (19, 7) (N = 40) and 0, 2, 3, 6 of m3 (21, 7) (N = 42), from `research/dreg_closure_scale_20261016/draws/`. Soundness: the 7 satisfiable draws of the same cells.
- Host: 4-core, 15 GB container shared with other jobs, not isolated; times are practicality notes only.

Measured

All 15 runs exited 0 and stopped at the 6.3 GB pivot cap with no verdict (measured):

cell	N	rows	columns	rows inserted at cap	engine secs
(19, 7), 8 draws	40	218,848	619,720	102,464–104,640 (46.8–47.8%)	253–292
(21, 7), 7 draws	42	278,978	778,324	77,312–79,488 (27.7–28.5%)	242–284

Registered outcome: 8 of 8 unmeasured. Not refuted and refuted are both unsupported. Controls: positive 20 of 20 correct (degree 5 and 6), soundness 8 of 8 not refuted, 0 failures. Size correction: the brief's 34,482 rows is 218,918 rows measured (17.0 GB dense).



blue N=40, red N=42; light bars and * = satisfiable draws (soundness). Source: runs/*.jsonl, run-20261018T2104Z. n=15 runs.

Figure 1: Rows inserted when the 6.3 GB pivot cap stopped each run, % of dense rows. Recorded values, n = 15 runs, run-20261018T2104Z. Light bars: satisfiable draws.

Missing

No degree-5 verdict for any registered draw; no derived slope point is available. Records: `runs/registered-*.jsonl`, `runs/soundness-*.jsonl`, `runs/ctl-*.jsonl`, `runs/queue.log`.